Lösung: komplexe Wurzeln

a) $z^4 = a = 81\, e^{i\pi/4} = 81 \cdot e^{i(\pi/4 + k\cdot 2\pi)} \quad \forall k \in \mathbb{Z}$

$$z_k = \sqrt[4]{81}\; e^{i(\pi/4 + k\cdot 2\pi)/4} = 3 \cdot e^{i(\pi/16 + k\pi/2)}$$

$$\left.\begin{aligned} z_0 &= 3 \cdot e^{i\pi/16} \\ z_1 &= 3 \cdot e^{i 9\pi/16} \\ z_2 &= 3 \cdot e^{i 17/16\,\pi} \\ z_3 &= 3 \cdot e^{i 25/16\,\pi} \end{aligned}\right\} \text{ 4 komplexe Wurzeln}$$

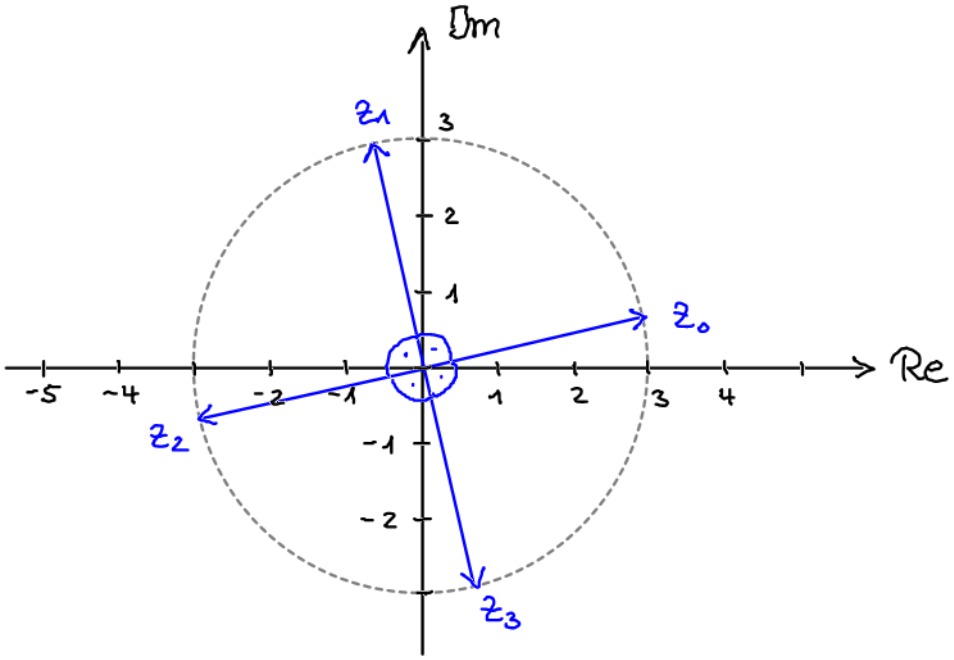
$$z_4 = 3 \cdot e^{i 33/16\,\pi} = 3 \cdot e^{i\pi/16} \cdot e^{i2\pi} = 3\, e^{i\pi/16} = z_0$$

$z_5 = \ldots = z_1$

$z_6 = \ldots = z_2$

$\vdots$

$z_n = \ldots = z_{n-4}$



b) $z^3 = a = \frac{-1+i}{\sqrt{2}} = -\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}$

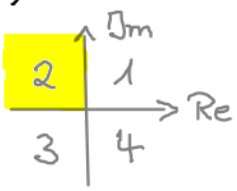
Überführe a in die Exponentialdarstellung

Betrag: $|a| = \sqrt{\left(-\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = \sqrt{\frac{1}{2}+\frac{1}{2}} = \sqrt{1} = 1$

Winkel: $\arg(a) = \arctan\left(\frac{-1/\sqrt{2}}{1/\sqrt{2}}\right) = \arctan(-1)$

$= -\pi/4 + \pi = \frac{3}{4}\pi \quad (\hat{=} 135°)$

↑ ..., da in 2. Quadrant



$\Leftrightarrow \quad a = e^{i\frac{3}{4}\pi} = e^{i(\frac{3}{4}\pi + 2\pi k)}$ mit $k \in \mathbb{Z}$

$\Leftrightarrow \quad z_k = e^{i(\frac{3}{4}\pi + 2\pi k)/3} = e^{i(\frac{\pi}{4} + \frac{2}{3}\pi \cdot k)}$

$$\left.\begin{aligned} z_0 &= e^{i\pi/4} \\ z_1 &= e^{i(\frac{\pi}{4}+\frac{2}{3}\pi)} = e^{i\frac{11}{12}\pi} \\ z_2 &= e^{i(\frac{\pi}{4}+\frac{4}{3}\pi)} = e^{i\frac{19}{12}\pi} \end{aligned}\right\} \text{ 3 komplexe Wurzeln}$$

$z_3 = e^{i(\frac{\pi}{4}+\frac{6}{3}\pi)} = e^{i\pi/4} \cdot e^{i2\pi} = e^{i\pi/4} = z_0$

$z_4 = \ldots = z_1$

$z_5 = \ldots = z_2$

$\vdots$

$z_n = \ldots = z_{n-3}$

